

# **LINUX**

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## **Topic Name:**

The main aim of this lab session is to provide hands-on experience on

- Getting Help
- Basic Commands
- Navigation
- File System
- Simple shell script
- Globbing
- Extended Globbing
- Filters -l (head,tail,wc)

## 1. Getting Help

| Task                                     | Command Name | Syntax        | Example | Screenshots                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|--------------|---------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To get manual page for the known command | man          | man <command> | man ls  |  <p>LS(1) User Commands LS(1)</p> <p><b>NAME</b></p> <p>ls - list directory contents</p> <p><b>SYNOPSIS</b></p> <p>ls [OPTION]... [FILE]...</p> <p><b>DESCRIPTION</b></p> <p>List information about the FILES (the current directory by default). Sort entries alphabetically if none of <code>-ctuvSUX</code> nor <code>--sort</code> is specified.</p> <p>Mandatory arguments to long options are mandatory for short options too.</p> <p><b>-a, --all</b><br/>do not ignore entries starting with <code>.</code></p> <p><b>-A, --almost-all</b><br/>do not list implied <code>.</code> and <code>..</code></p> <p><b>-author</b><br/>with <code>-l</code>, print the author of each file</p> <p><b>-b, --escape</b><br/>print C-style escapes for nongraphic characters</p> <p><b>--block-size=SIZE</b><br/>with <code>-l</code>, scale sizes by SIZE when printing them; e.g., <code>'--block-size=M'</code>; see SIZE format below</p> <p><b>-B, --ignore-backups</b><br/>do not list implied entries ending with <code>~</code></p> <p><b>-c</b> with <code>-lt</code>: sort by, and show, ctime (time of last change of file status information); with <code>-l</code>: show ctime and sort by name; otherwise: sort by ctime, newest first</p> <p><b>-C</b> list entries by columns</p> <p><b>--color[=WHEN]</b><br/>color the output WHEN; more info below</p> <p><b>-d, --directory</b><br/>list directories themselves, not their contents</p> <p><b>-D, --dired</b><br/>generate output designed for Emacs' dired mode</p> <p><b>-f</b> same as <code>-a -U</code></p> <p><b>-F, --classify[=WHEN]</b><br/>append indicator (one of <code>*/=&gt;@ </code>) to entries WHEN</p> <p>Manual page ls(1) line 1 (press h for help or q to quit)</p> |

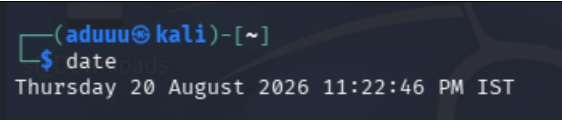
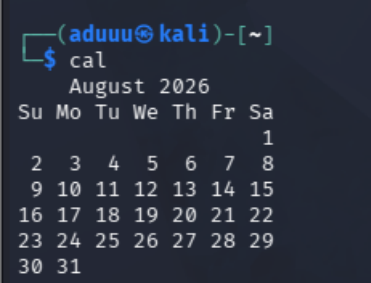
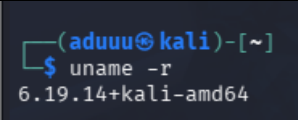
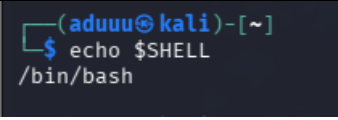
|                                            |                  |                   |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|------------------|-------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To get manual page for the unknown command | apropos / man -k | apropos <keyword> | apropos copy | <pre> (duuu@kali)-[~] └─\$ apropos copy Clone (3m) - recursively copy Perl datatypes bcopy (3) - copy byte sequence cifsd (8) - convert and copy a file over SMB COPY (7) - copy data between a file and a table copy_file_range (2) - copy a range of data from one file to another copyfilerange (1) - copy range(s) from source to destination file copysign (3) - copy sign of a number copysignf (3) - copy sign of a number copysignl (3) - copy sign of a number cp (1) - copy files and directories cpio (1) - copy files to and from archives dd (1) - convert and copy a file debconf-copydb (1) - copy a debconf database docker-container-cp (1) - Copy files/folders between a container and the local filesystem docker-cp (1) - Copy files/folders between a container and the local filesystem dvcopy (1) - produce modified copy of DVI file getunwind (2) - copy the unwind data to caller's buffer getutmp (3) - copy utmp structure to utmpx, and vice versa getutmpx (3) - copy utmp structure to utmpx, and vice versa git-checkout-index (1) - Copy files from the index to the working tree i866-w64-mingw32-objcopy (1) - copy and translate object files install (1) - copy files and set attributes llvm-dwarftool-21 (1) - A tool to copy and manipulate debug info llvm-objcopy-21 (1) - object copying and editing tool mariadb-hotcopy (1) - a database backup program (mariadb-hotcopy is now a symlink to mariadb-hotcopy) memccpy (3) - copy memory area memcpy (3) - copy memory area memmove (3) - copy memory area memcpy (3) - copy memory area ntfscp (8) - copy file to an NTFS volume. objcopy (1) - copy and translate object files odvcopy (1) - produce modified copy of DVI file pantopam (1) - copy PAM image pgmtopgm (1) - copy PGM image pmtopmm (1) - copy a PWM image pmtoppm (1) - copy PPM image ptrepack (1) - Copy any PyTables Leaf, Group or complete subtree into another file. rsync (1) - a fast, versatile, remote (and local) file-copying tool scp (1) - OpenSSH secure file copy scp1 (1) - secure copy (remote file copy program) ssh-copy-id (1) - use locally available keys to authorise logins on a remote machine stpcpy (3) - string return-offset-pointer copy </pre> |
| To know the source file binary             | type -a          | type -a <command> | type -a ls   | <pre> (duuu@kali)-[~] └─\$ type -a ls ls is aliased to `ls --color=auto' ls is /usr/bin/ls ls is /bin/ls </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|---------------------------------------------|-------|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To know the path of the command             | which | which <command> | which ls | <pre>(aduuu@kali)-[~] \$ which ls /usr/bin/ls</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| To know the command is external or internal | type  | type <command>  | type cd  | <pre>(aduuu@kali)-[~] \$ type cd cd is a shell builtin</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| To get help for the internal command        | help  | help <command>  | help cd  | <pre>(aduuu@kali)-[~] \$ help cd cd: cd [-L [-P [-e]] [-@]] [dir] Change the shell working directory.  Change the current directory to DIR. The default DIR is the value of the HOME shell variable. If DIR is "-", it is converted to \$OLDPWD.  The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory. If DIR begins with a slash (/), then CDPATH is not used.  If the directory is not found, and the shell option 'cdable_vars' is set, the word is assumed to be a variable name. If that variable has a value, its value is used for DIR.  Options:   -L      force symbolic links to be followed: resolve symbolic           links in DIR after processing instances of '..'   -P      use the physical directory structure without following           symbolic links: resolve symbolic links in DIR before           processing instances of '..'   -e      if the -P option is supplied, and the current working           directory cannot be determined successfully, exit with           a non-zero status   -@      on systems that support it, present a file with extended           attributes as a directory containing the file attributes  The default is to follow symbolic links, as if '-L' were specified. '..' is processed by removing the immediately previous pathname component back to a slash or the beginning of DIR.  Exit Status: Returns 0 if the directory is changed, and if \$PWD is set successfully when -P is used; non-zero otherwise.</pre> |

|                           |          |             |             |                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|----------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To list out bash commands | compngen | compngen -b | compngen -b | <pre> (adiuuu@kali)-[~] \$ compngen -b  : [ alias bg bind break builtin caller cd command compngen complete compropt continue declare dirs disown echo enable eval exec exit export false fc fg getopts hash help history jobs kill let local logout mapfile popd printf pushd pwd read readarray readonly return set shift shopt source </pre> |
|---------------------------|----------|-------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                  |        |                  |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|--------|------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To know the usage of the command | --help | <command> --help | ls --help | <pre> (adi@kali)~\$ ls --help Usage: ls [OPTION]... [FILE]... List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.  Mandatory arguments to long options are mandatory for short options too. -a, --all     do not ignore entries starting with . -A, --almost-all     do not list implied . and .. --author     with -l, print the author of each file -b, --escape     print C-style escapes for nongraphic characters --block-size=SIZE     with -l, scale sizes by SIZE when printing them;     e.g., '--block-size=M'; see SIZE format below -B, --ignore-backups     do not list implied entries ending with ~ -c     with -lt: sort by, and show, ctime     (time of last change of file status information);     with -l: show ctime and sort by name;     otherwise: sort by ctime, newest first -C     list entries by columns --color[=WHEN]     color the output WHEN; more info below -d, --directory     list directories themselves, not their contents -D, --dired     generate output designed for Emacs' dired mode -f     same as -a -U -F, --classify[=WHEN]     append indicator (one of */=&gt;@ ) to entries WHEN --file-type     like -F, except do not append '*' --format=WORD     across, horizontal (-x), commas (-m), long (-l),     single-column (-1), verbose (-l), vertical (-C) --full-time     like -l --time-style=full-iso -g     like -l, but do not list owner --group-directories-first     group directories before files -G, --no-group     in a long listing, don't print group names -h, --human-readable </pre> |
|----------------------------------|--------|------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2. Basic Commands

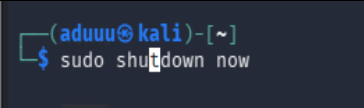
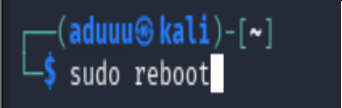
| Task                    | Command Name | Syntax       | Example     | Screenshots                                                                           |
|-------------------------|--------------|--------------|-------------|---------------------------------------------------------------------------------------|
| To know today's date    | date         | date         | Date        |    |
| To print calendar       | cal          | cal          | Cal         |    |
| To print kernel version | uname        | uname -r     | uname -r    |   |
| echo                    | echo         | echo \$SHELL | echo\$SHELL |  |

|                                   |         |                        |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|-----------------------------------|---------|------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| To print currently logged in user | whoami  | whoami                 | whoami             | <pre>(aduuu@kali)-[~] \$ whoami aduuu</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
| To create shortcut for command    | alias   | alias <name>=<command> | alias list='ls -l' | <pre>(aduuu@kali)-[~] \$ alias list='ls -l'  (aduuu@kali)-[~] \$ list total 64 drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Desktop drwxrwxr-x 2 aduuu aduuu 4096 Aug 12 21:54 document drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Documents drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Downloads -rw-rw-r-- 1 aduuu aduuu 0 Aug 20 23:13 file1.txt -rw-rw-r-- 1 aduuu aduuu 0 Aug 20 23:12 file.txt drwxrwxr-x 2 aduuu aduuu 4096 Aug 12 21:49 linux drwxrwxr-x 2 aduuu aduuu 4096 Aug 12 21:51 Linux drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Music -rw-r--r-- 1 aduuu aduuu 0 Aug 12 21:51 notes.txt drwxr-xr-x 2 aduuu aduuu 4096 Aug 18 11:53 Pictures drwxrwxr-x 2 aduuu aduuu 4096 Aug 13 00:11 practice drwxrwxr-x 2 aduuu aduuu 4096 Aug 13 00:12 Practice drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Projects drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Public drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Templates -rwxrwxr-x 1 aduuu aduuu 162 Aug 11 16:18 tool.sh -rwxrwxr-x 1 aduuu aduuu 28 Aug 10 22:42 update-system.sh drwxr-xr-x 2 aduuu aduuu 4096 Aug 10 00:08 Videos</pre> |  |
| To delete shortcut                | unalias | unalias <name>         | unalias list       | <pre>(aduuu@kali)-[~] \$ unalias list  (aduuu@kali)-[~] \$ list Command 'list' not found, did you mean:   command 'last' from deb wtmpdb   command 'flist' from deb nmh   command 'mlist' from deb mblaze   command 'hist' from deb loki   command 'dist' from deb nmh   command 'lift' from deb lift   command 'gist' from deb yorick Try: sudo apt install &lt;deb name&gt;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |

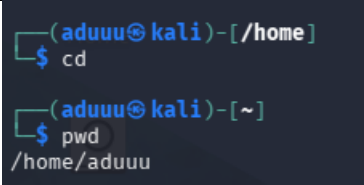
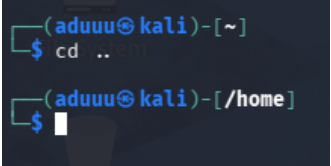
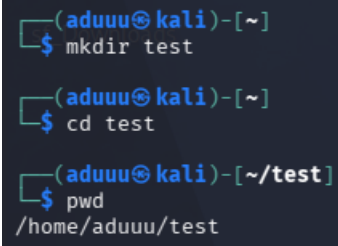


|                                     |             |                    |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|-------------|--------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To change the timestamp of the file | touch       | touch <filename>   | touch file.txt  | <pre>(aduuu@kali)-[~] \$ touch file.txt  (aduuu@kali)-[~] \$</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| To clear the screen                 | clear       | clear              | Clear           | <pre>(aduuu@kali)-[~] \$ clear</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| To create empty files               | touch       | touch <filename>   | touch file1.txt | <pre>(aduuu@kali)-[~] \$ touch file1.txt  (aduuu@kali)-[~] \$</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| To know disk usage                  | du          | du -sh <directory> | du -sh .        | <pre>(aduuu@kali)-[~] \$ du -sh . 151M .</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| To know free space in the system    | df          | df -h              | df -h           | <pre>(aduuu@kali)-[~] \$ df -h Filesystem      Size  Used Avail Use% Mounted on udev            857M   0 857M   0% /dev tmpfs           193M 1004K 192M   1% /run /dev/sda1        35G   16G  17G  49% / tmpfs           961M   8.0K 961M   1% /dev/shm tmpfs           961M   8.0K 961M   1% /tmp Downloads       230G  173G   58G  76% /media/sf_Downloads none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service tmpfs           193M   3.0M 190M   2% /run/user/1000 none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service</pre> |
| To know about the Linux release     | lsb_release | lsb_release -a     | lsb_release -a  | <pre>(aduuu@kali)-[~] \$ lsb_release -a No LSB modules are available. Distributor ID: Kali Description:    Kali GNU/Linux Rolling Release:        2026.2 Codename:       kali-rolling</pre>                                                                                                                                                                                                                                                                                                                                                                                 |

|                                      |        |                     |                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|--------|---------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| To show login history                | last   | last                | last                | <pre>(aduuu@kali)-[~] \$ last aduuu    tty7          :0                Thu Aug 20 20:36 - still logged in aduuu    tty7          :0                Tue Aug 18 12:59 - still logged in aduuu    tty7          :0                Tue Aug 18 11:35 - still logged in aduuu    tty7          :0                Mon Aug 17 20:27 - still logged in aduuu    tty7          :0                Sat Aug 15 09:16 - still logged in aduuu    tty7          :0                Sat Aug 15 01:40 - still logged in aduuu    tty7          :0                Wed Aug 12 21:47 - still logged in aduuu    tty7          :0                Tue Aug 11 14:45 - still logged in aduuu    tty7          :0                Mon Aug 10 23:46 - still logged in aduuu    tty7          :0                Mon Aug 10 00:08 - 23:25 (23:17)  wtm pdb begins Mon Aug 10 00:08:08 2026</pre> |
| To check the current user privileges | id     | id                  | id                  | <pre>(aduuu@kali)-[~] \$ id uid=1000(aduuu) gid=1000(aduuu) groups=1000(aduuu),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plu gdev),100(users),101(netdev),102(scanner),968(vboxsf),979(bluetooth),999(lpadmin)</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| To find about the Linux release      | cat    | cat /etc/os-release | cat /etc/os-release | <pre>(aduuu@kali)-[~] \$ cat /etc/os-release PRETTY_NAME="Kali GNU/Linux Rolling" NAME="Kali GNU/Linux" VERSION_ID="2026.2" VERSION="2026.2" VERSION_CODENAME=kali-rolling ID=kali ID_LIKE=debian HOME_URL="https://www.kali.org/" SUPPORT_URL="https://forums.kali.org/" BUG_REPORT_URL="https://bugs.kali.org/" ANSI_COLOR="1;31"</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| To check system uptime               | uptime | uptime              | Uptime              | <pre>(aduuu@kali)-[~] \$ uptime 23:22:27 up 2:47, 1 user, load average: 0.33, 0.27, 0.26</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

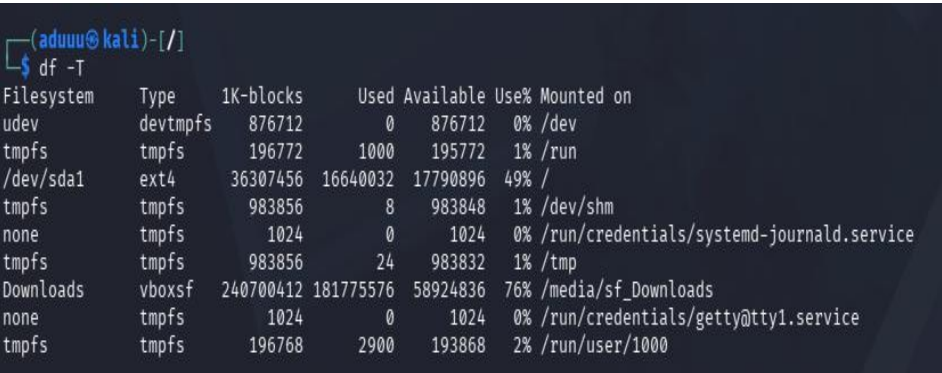
|             |          |                   |                   |                                                                                     |
|-------------|----------|-------------------|-------------------|-------------------------------------------------------------------------------------|
| To shutdown | shutdown | sudo shutdown now | sudo shutdown now |  |
| To restart  | reboot   | sudo reboot       | sudo reboot       |  |

### 3. Navigation

| Task                                | Command | Syntax        | Screenshots                                                                           |
|-------------------------------------|---------|---------------|---------------------------------------------------------------------------------------|
| To navigate home directory          | cd      | cd            |    |
| To navigate to the parent directory | cd      | cd ..         |   |
| To navigate to the child directory  | cd      | cd foldername |  |

|                                      |      |                 |                                                                     |
|--------------------------------------|------|-----------------|---------------------------------------------------------------------|
| Alternate command to cd              | pusd | pushd directory | <pre>(aduuu@kali)-[~/test] \$ pushd .. ~ ~/test</pre>               |
| To go back to the previous directory | cd-  | cd-             | <pre>(aduuu@kali)-[~] \$ cd - /home/aduuu/test</pre>                |
| To go to the root directory          | cd/  | cd/             | <pre>(aduuu@kali)-[~/test] \$ cd /  (aduuu@kali)-[/] \$ pwd /</pre> |

#### 4. File System

| Task                            | Syntax | Command | Screenshots                                                                          |
|---------------------------------|--------|---------|--------------------------------------------------------------------------------------|
| How to identify the file system | df -T  | df -T   |  |

## 5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use globbing patterns wherever possible, rather than specifying filenames individually.

```
(aduuu@kali)-[~]
$ mkdir practice1

(aduuu@kali)-[~]
$ cd practice1

(aduuu@kali)-[~/practice1]
$ pwd
/home/aduuu/practice1

(aduuu@kali)-[~/practice1]
$ ls -ld~
ls: invalid option -- '~'
Try 'ls --help' for more information.

(aduuu@kali)-[~/practice1]
$ ls -ld ~
drwx----- 26 aduuu aduuu 4096 Aug 21 00:52 /home/aduuu

(aduuu@kali)-[~/practice1]
$ pwd
/home/aduuu/practice1

(aduuu@kali)-[~/practice1]
$ touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.txt file123 backup1.html backup2.txt

(aduuu@kali)-[~/practice1]
$ ls
access.log  admin.htm  backup1.html  backup2.txt  error.log  file123  index.html  login.html  report.txt  user1.txt  user2.txt
```

- a. List all files ending in .html.

```
(aduuu@kali) - [~/practice1]
$ ls *.html
backup1.html  index.html  login.html
```

- b. List all files other than .html files using extended globbing.

```
(aduuu@kali) - [~/practice1]
$ shopt -s extglob

(aduuu@kali) - [~/practice1]
$ ls !(*.html)
access.log  admin.htm  backup2.txt  error.log  file123  report.txt  user1.txt  user2.txt
```

- c. List filenames containing at least one numeric digit.

```
(aduuu@kali) - [~/practice1]
$ ls *[0-9]*
backup1.html  backup2.txt  file123  user1.txt  user2.txt
```

- d. List files whose names begin with user followed by a single digit.

```
(aduuu@kali) - [~/practice1]
$ ls user?.*
user1.txt  user2.txt
```

- e. List files ending in either .txt or .log using a single extended glob pattern.

```
(aduuu@kali) - [~/practice1]
$ ls *.@(txt|log)
access.log  backup2.txt  error.log  report.txt  user1.txt  user2.txt
```

- f. List files that do not end in .txt or .log.

```
(aduuu@kali) - [~/practice1]
$ ls !(*.txt|*.log)
admin.htm  backup1.html  file123  index.html  login.html
```

- g. List files beginning with either access or error using extended globbing.

```
(aduuu@kali) - [~/practice1]
$ ls @(access|error).log
access.log  error.log
```

- h. List .html files whose filenames contain a numeric digit.

```
(aduuu@kali) - [~/practice1]
$ ls *[0-9]*.html
backup1.html
```

## 6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

Select all files except notes.txt.

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately

```
(aduuu@kali) - [~/practice1]
$ mkdir practice2

(aduuu@kali) - [~/practice1]
$ cd practice2

(aduuu@kali) - [~/practice1/practice2]
$ touch access.log error.log auth.log auth.log.1 auth.log.2 system.log notes.txt

(aduuu@kali) - [~/practice1/practice2]
$ shopt -s extglob

(aduuu@kali) - [~/practice1/practice2]
$ ls auth.log.@(1|2)
auth.log.1  auth.log.2

(aduuu@kali) - [~/practice1/practice2]
$ ls @(access|error|auth).log
access.log  auth.log  error.log

(aduuu@kali) - [~/practice1/practice2]
$ ls !(notes.txt)
access.log  auth.log  auth.log.1  auth.log.2  error.log  system.log

(aduuu@kali) - [~/practice1/practice2]
$ ls auth.log.@(1|2)
auth.log.1  auth.log.2
```



b. Given the files:

malware1.log, malware2.log, malwareA.log, malware\_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
(aduuu@kali)-[~/practice1/practice2]
$ touch malware1.log malware2.log malwareA.log malware_report.txt normal.log

(aduuu@kali)-[~/practice1/practice2]
$ shopt -s extglob

(aduuu@kali)-[~/practice1/practice2]
$ ls malware*[0-9]*.log
malware1.log  malware2.log

(aduuu@kali)-[~/practice1/practice2]
$ ls malware*[0-9]*.log
malware1.log  malware2.log
```

c. Given:

failed\_login.log, successful\_login.log, login\_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed\_login or successful\_login.

Construct a pattern that lists all .log files except firewall.log.

```
(aduuu@kali)-[~/practice1/practice2]
$ touch failed_login.log successful_login.log login_backup.log login.txt firewall.log

(aduuu@kali)-[~/practice1/practice2]
$ shopt -s extglob

(aduuu@kali)-[~/practice1/practice2]
$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log

(aduuu@kali)-[~/practice1/practice2]
$ ls !("firewall.log")
failed_login.log  login_backup.log  login.txt  successful_login.log
```

## 7. Filters

```
(aduuu@kali)-[~]  
$ mkdir practice3  
  
(aduuu@kali)-[~]  
$ cd practice3  
  
(aduuu@kali)-[~/practice3]  
$ seq 1 100 | awk '{print "login entry "$1}' > login.log  
  
(aduuu@kali)-[~/practice3]  
$ seq 1 100 | awk '{print "access entry "$1}' > access.log  
  
(aduuu@kali)-[~/practice3]  
$ seq 1 500 | awk '{print "security event "$1}' > security.log  
  
(aduuu@kali)-[~/practice3]  
$ seq 1 500 | awk '{print "incident event "$1}' > incident.log
```

a. Display the first 10 entries of login.log.

```
(aduuu@kali)-[~/practice3]  
$ head -10 login.log  
login entry 1  
login entry 2  
login entry 3  
login entry 4  
login entry 5  
login entry 6  
login entry 7  
login entry 8  
login entry 9  
login entry 10
```

b. Display the first 5 entries of access.log.

```
(aduuu@kali)-[~/practice3]  
$ head -5 access.log  
access entry 1  
access entry 2  
access entry 3  
access entry 4  
access entry 5
```

- c. Display the first 15 security events from security.log.

```
(aduuu@kali)-[~/practice3]
$ head -15 security.log
security event 1
security event 2
security event 3
security event 4
security event 5
security event 6
security event 7
security event 8
security event 9
security event 10
security event 11
security event 12
security event 13
security event 14
security event 15
```

- d. Display the first 20 events recorded in incident.log.

```
(aduuu@kali)-[~/practice3]
$ head -20 incident.log
incident event 1
incident event 2
incident event 3
incident event 4
incident event 5
incident event 6
incident event 7
incident event 8
incident event 9
incident event 10
incident event 11
incident event 12
incident event 13
incident event 14
incident event 15
incident event 16
incident event 17
incident event 18
incident event 19
incident event 20
```

- e. Display only the first entry in login.log.

```
(aduuu@kali)-[~/practice3]  
$ head -1 login.log  
login entry 1
```

- f. Display the first 25 web requests recorded in access.log.

```
(aduuu@kali)-[~/practice3]  
$ head -25 access.log  
access entry 1  
access entry 2  
access entry 3  
access entry 4  
access entry 5  
access entry 6  
access entry 7  
access entry 8  
access entry 9  
access entry 10  
access entry 11  
access entry 12  
access entry 13  
access entry 14  
access entry 15  
access entry 16  
access entry 17  
access entry 18  
access entry 19  
access entry 20  
access entry 21  
access entry 22  
access entry 23  
access entry 24  
access entry 25
```

- g. Display the last 10 entries of login.log.

```
(aduuu@kali)-[~/practice3]  
$ tail -10 login.log  
login entry 91  
login entry 92  
login entry 93  
login entry 94  
login entry 95  
login entry 96  
login entry 97  
login entry 98  
login entry 99  
login entry 100
```

- h. Display the last 5 entries of access.log.

```
(aduuu@kali)-[~/practice3]  
$ tail -5 access.log  
access entry 96  
access entry 97  
access entry 98  
access entry 99  
access entry 100
```

- i. Display the 20 most recent security events from security.log.

```
(aduuu@kali)-[~/practice3]  
$ tail -20 security.log  
security event 481  
security event 482  
security event 483  
security event 484  
security event 485  
security event 486  
security event 487  
security event 488  
security event 489  
security event 490  
security event 491  
security event 492  
security event 493  
security event 494  
security event 495  
security event 496  
security event 497  
security event 498  
security event 499  
security event 500
```

- j. Display the 25 most recent events from incident.log.

```
(aduuu@kali)~[/practice3]
$ tail -25 incident.log
incident event 476
incident event 477
incident event 478
incident event 479
incident event 480
incident event 481
incident event 482
incident event 483
incident event 484
incident event 485
incident event 486
incident event 487
incident event 488
incident event 489
incident event 490
incident event 491
incident event 492
incident event 493
incident event 494
incident event 495
incident event 496
incident event 497
incident event 498
incident event 499
incident event 500
```

- k. Display only the last entry in security.log.

```
(aduuu@kali)~[/practice3]
$ tail -1 security.log
security event 500
```



I. Display the last 50 events recorded in incident.log.

```
(aduuu@kali)~[/practice3]
$ tail -50 incident.log
incident event 451
incident event 452
incident event 453
incident event 454
incident event 455
incident event 456
incident event 457
incident event 458
incident event 459
incident event 460
incident event 461
incident event 462
incident event 463
incident event 464
incident event 465
incident event 466
incident event 467
incident event 468
incident event 469
incident event 470
incident event 471
incident event 472
incident event 473
incident event 474
incident event 475
incident event 476
incident event 477
incident event 478
incident event 479
incident event 480
incident event 481
incident event 482
incident event 483
incident event 484
incident event 485
incident event 486
incident event 487
incident event 488
incident event 489
incident event 490
incident event 491
incident event 492
incident event 493
incident event 494
incident event 495
```

m. Display the contents of login.log starting from line 41 until the end of the file.

```
(aduuu@kali)-[~/practice3]
$ tail -n +41 login.log
login entry 41
login entry 42
login entry 43
login entry 44
login entry 45
login entry 46
login entry 47
login entry 48
login entry 49
login entry 50
login entry 51
login entry 52
login entry 53
login entry 54
login entry 55
login entry 56
login entry 57
login entry 58
login entry 59
login entry 60
login entry 61
login entry 62
login entry 63
login entry 64
login entry 65
login entry 66
login entry 67
login entry 68
login entry 69
login entry 70
login entry 71
login entry 72
login entry 73
login entry 74
login entry 75
login entry 76
login entry 77
login entry 78
login entry 79
login entry 80
login entry 81
login entry 82
login entry 83
```

- n. Display the contents of security.log starting from line 101 until the end of the file.

```
(aduuu@kali)-[~/practice3]
$ tail -n +101 security.log
security event 101
security event 102
security event 103
security event 104
security event 105
security event 106
security event 107
security event 108
security event 109
security event 110
security event 111
security event 112
security event 113
security event 114
security event 115
security event 116
security event 117
security event 118
security event 119
security event 120
security event 121
security event 122
security event 123
security event 124
security event 125
security event 126
security event 127
security event 128
security event 129
security event 130
security event 131
security event 132
security event 133
security event 134
security event 135
security event 136
security event 137
security event 138
```

- o. Find the total number of lines in login.log.

```
(aduuu@kali)-[~/practice3]  
$ wc -l login.log  
100 login.log
```

- p. Find the total number of lines in access.log.

```
(aduuu@kali)-[~/practice3]  
$ wc -l access.log  
100 access.log
```

- q. Find the total number of security events in security.log.

```
(aduuu@kali)-[~/practice3]  
$ wc -l security.log  
500 security.log
```

- r. Determine the total number of events recorded in incident.log.

```
(aduuu@kali)-[~/practice3]  
$ wc -l incident.log  
500 incident.log
```

- s. Count the number of words in login.log.

```
(aduuu@kali)-[~/practice3]  
$ wc -w login.log  
300 login.log
```

- t. Count the number of bytes in security.log.

```
(aduuu@kali)~/practice3  
$ wc -c security.log  
9392 security.log
```

- u. Display the line, word, and byte counts of access.log.

```
(aduuu@kali)~/practice3  
$ wc access.log  
100 300 1592 access.log
```

- v. Find the line counts of all four .log files using a single wc command.

```
(aduuu@kali)~/practice3  
$ wc -l login.log access.log security.log incident.log  
100 login.log  
100 access.log  
500 security.log  
500 incident.log  
1200 total
```

- w. Display the first 20 entries of login.log and, from those entries, display only the last 5.

```
(aduuu@kali)~/practice3  
$ head -20 login.log | tail -5  
login entry 16  
login entry 17  
login entry 18  
login entry 19  
login entry 20
```

- x. Display the last 20 entries of login.log and, from those entries, display only the first 5.

```
(aduuu@kali)~/practice3  
$ tail -20 login.log | head -5  
login entry 81  
login entry 82  
login entry 83  
login entry 84  
login entry 85
```

- y. Display the first 30 entries of access.log and then display only the last 10 entries from that output.

```
(aduuu@kali)-[~/practice3]
$ head -30 access.log | tail -10
access entry 21
access entry 22
access entry 23
access entry 24
access entry 25
access entry 26
access entry 27
access entry 28
access entry 29
access entry 30
```

- z. Display the last 30 entries of access.log and then display only the first 10 entries.

```
(aduuu@kali)-[~/practice3]
$ tail -30 access.log | head -10
access entry 71
access entry 72
access entry 73
access entry 74
access entry 75
access entry 76
access entry 77
access entry 78
access entry 79
access entry 80
```

- aa. Display the first 50 events from security.log and then display the last 10 events from those 50.

```
(aduuu@kali)-[~/practice3]
$ head -50 security.log | tail -10
security event 41
security event 42
security event 43
security event 44
security event 45
security event 46
security event 47
security event 48
security event 49
security event 50
```

bb. Display the last 50 events from security.log and then display the first 10 events from those 50.

```
(aduuu@kali)-[~/practice3]
$ tail -50 security.log | head -10
security event 451
security event 452
security event 453
security event 454
security event 455
security event 456
security event 457
security event 458
security event 459
security event 460
```

cc. Display entries from line 11 to line 20 of login.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '11,20p' login.log
login entry 11
login entry 12
login entry 13
login entry 14
login entry 15
login entry 16
login entry 17
login entry 18
login entry 19
login entry 20
```

dd. Display entries from line 21 to line 30 of login.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '21,30p' login.log
login entry 21
login entry 22
login entry 23
login entry 24
login entry 25
login entry 26
login entry 27
login entry 28
login entry 29
login entry 30
```

ee. Display entries from line 11 to line 20 of access.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '11,20p' access.log
access entry 11
access entry 12
access entry 13
access entry 14
access entry 15
access entry 16
access entry 17
access entry 18
access entry 19
access entry 20
```

ff. Display entries from line 31 to line 40 of access.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '31,40p' access.log
access entry 31
access entry 32
access entry 33
access entry 34
access entry 35
access entry 36
access entry 37
access entry 38
access entry 39
access entry 40
```

gg. Display security events from line 21 to line 30 of security.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '21,30p' security.log
security event 21
security event 22
security event 23
security event 24
security event 25
security event 26
security event 27
security event 28
security event 29
security event 30
```



hh. Display security events from line 51 to line 60 of security.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '51,60p' security.log
security event 51
security event 52
security event 53
security event 54
security event 55
security event 56
security event 57
security event 58
security event 59
security event 60
```

ii. Display incident events from line 101 to line 110 of incident.log.

```
(aduuu@kali)-[~/practice3]
$ sed -n '101,110p' incident.log
incident event 101
incident event 102
incident event 103
incident event 104
incident event 105
incident event 106
incident event 107
incident event 108
incident event 109
incident event 110
```

jj. Display incident events from line 201 to line 220 of incident.log.

```
(aduuu@kali)~/practice3
$ sed -n '201,220p' incident.log
incident event 201
incident event 202
incident event 203
incident event 204
incident event 205
incident event 206
incident event 207
incident event 208
incident event 209
incident event 210
incident event 211
incident event 212
incident event 213
incident event 214
incident event 215
incident event 216
incident event 217
incident event 218
incident event 219
incident event 220
```

kk. Display incident events from line 491 to line 500.

```
(aduuu@kali)~/practice3
$ sed -n '491,500p' incident.log
incident event 491
incident event 492
incident event 493
incident event 494
incident event 495
incident event 496
incident event 497
incident event 498
incident event 499
incident event 500
```

ll. Take the first 20 entries of login.log and count how many lines are displayed.

```
(aduuu@kali)~/practice3  
$ head -20 login.log | wc -l  
20
```

mm. Take the last 15 entries of access.log and count how many lines are displayed.

```
(aduuu@kali)~/practice3  
$ tail -15 access.log | wc -l  
15
```

nn. Take the first 50 entries of security.log and count the number of words in those entries.

```
(aduuu@kali)~/practice3  
$ head -50 security.log | wc -w  
150
```

oo. Take the last 25 entries of security.log and count the number of lines.

```
(aduuu@kali)~/practice3  
$ tail -25 security.log | wc -l  
25
```

pp. Extract lines 21–30 from login.log and count the number of extracted lines.

```
(aduuu@kali)~/practice3  
$ sed -e '21,30p' login.log | wc -l  
110
```

qq. Extract lines 51–60 from access.log and count the number of extracted lines.

```
(aduuu@kali)~/practice3  
$ sed -n '51,60p' access.log | wc -l  
10
```

rr. Extract events 101–150 from incident.log and count how many events are displayed.

```
(aduuu@kali)~/practice3  
$ sed -n '101,150p' incident.log | wc -l  
50
```

ss. Display the first 100 events from incident.log, take the last 20 events from that output, and count the number of lines.

```
(aduuu@kali)~/practice3  
$ head -100 incident.log | tail -20 | wc -l  
20
```